

Kinetic Modeling of Cell Signaling

Module 1: Reaction networks as dynamical systems

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MSCBIO 2025 · Introduction to Bioinformatics Programming in Python

Fall 2026

Resources

- ▶ **BioNetGen** software, documentation, and tutorials:
bionetgen.org
- ▶ **VS Code extension:** search the Extensions panel for *BioNetGen*
- ▶ **BNG Playground:** write and simulate models in your browser, no installation
ruleworld.github.io/bngplayground
- ▶ **bngsim:** fast ODE and stochastic simulation from Python
bngsim.readthedocs.io · `pip install bngsim bionetgen`
- ▶ **Course repository:** models, notebooks, slides and handouts
github.com/jrfaeder/RBM_Intro_BioNetGen



Course repository



BNG Playground

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4. encode the same network in the **BioNetGen language** (BNGL), simulate it, and interpret the results
5. explain how a **negative feedback** loop shapes the dynamics of a signaling response

Part 1

Why model reaction networks?

Reaction networks describe how molecule populations change

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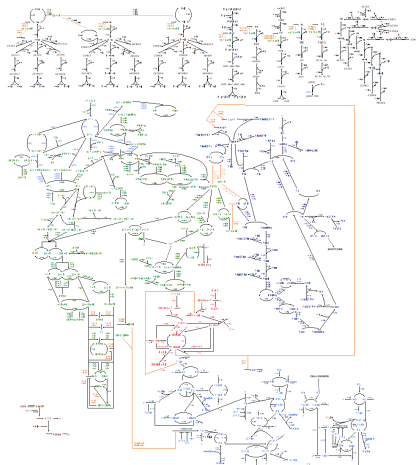
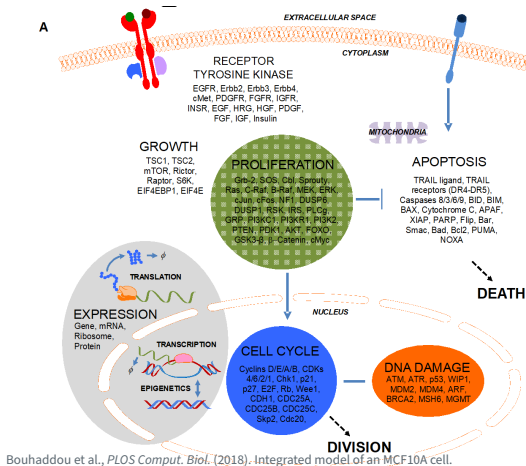
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What a model gives us

- ▶ A precise statement of a mechanism
- ▶ Predictions of dynamics we cannot easily measure
- ▶ A way to test whether a mechanism can explain data
- ▶ Hypotheses for the next experiment

Large models are built from simple reaction steps

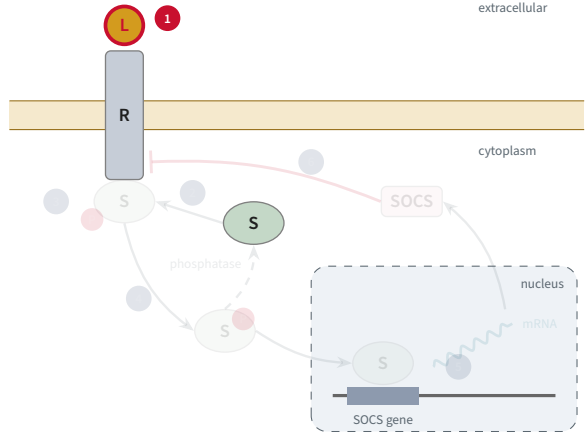


Part 2

A cytokine signaling circuit

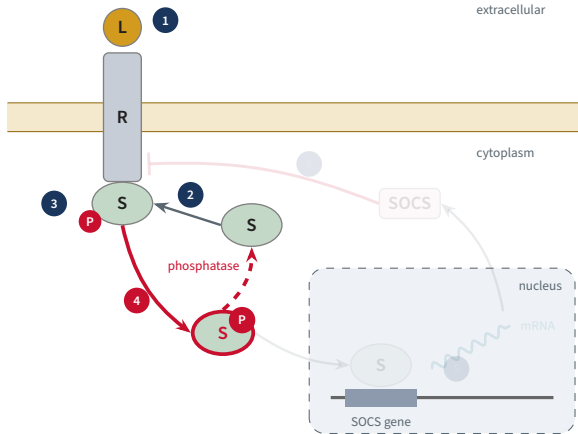
Cytokines signal through receptors, STATs, and SOCS feedback

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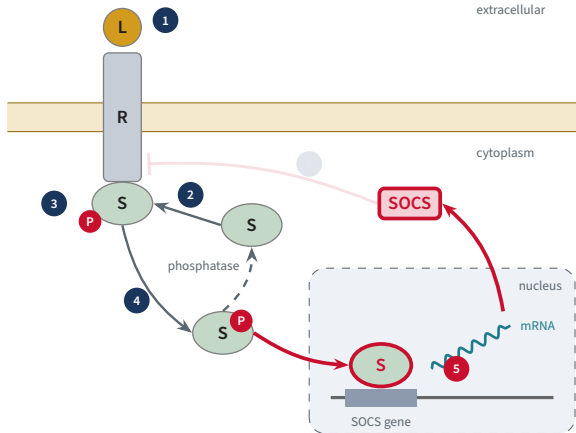
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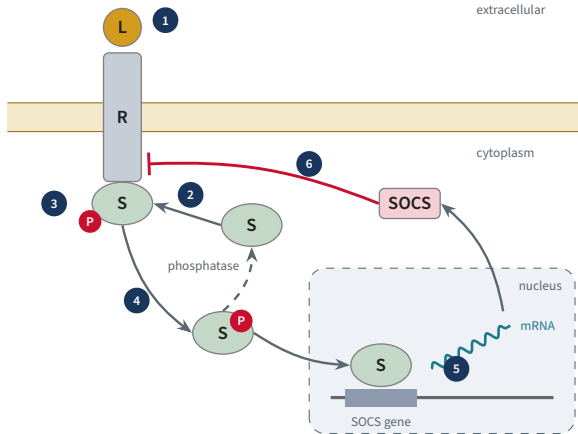
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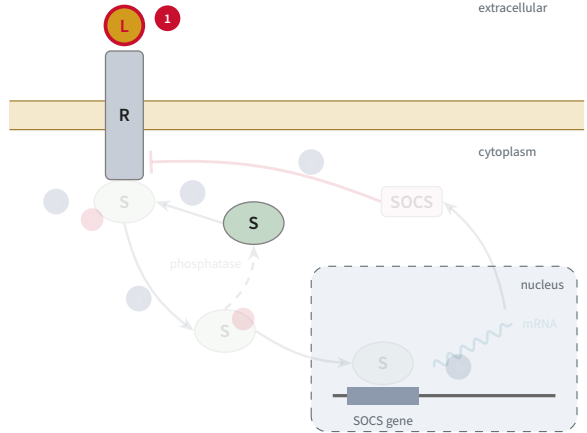
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- ▶ SOCS binds the receptor and **shuts off further STAT activation**: negative feedback.



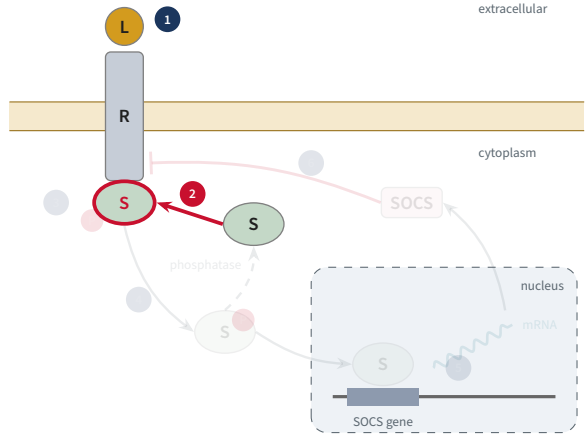
We will build the model one reaction step at a time

1. Ligand-receptor binding



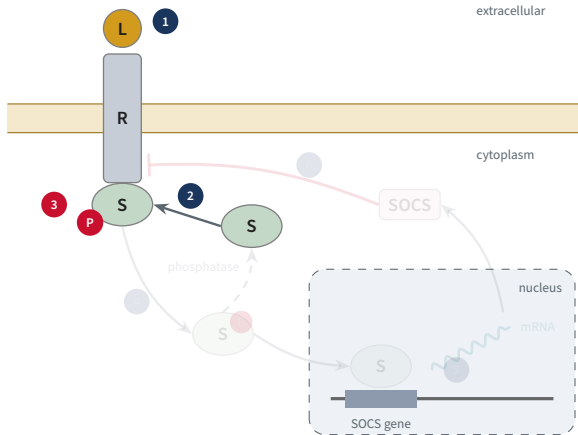
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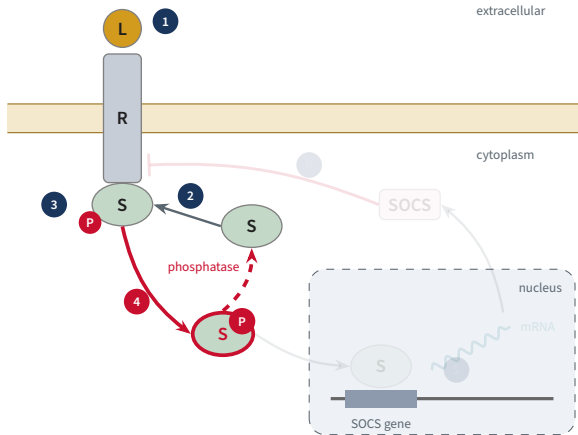
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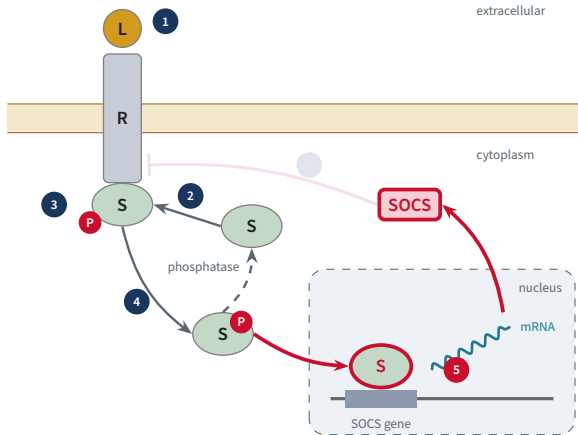
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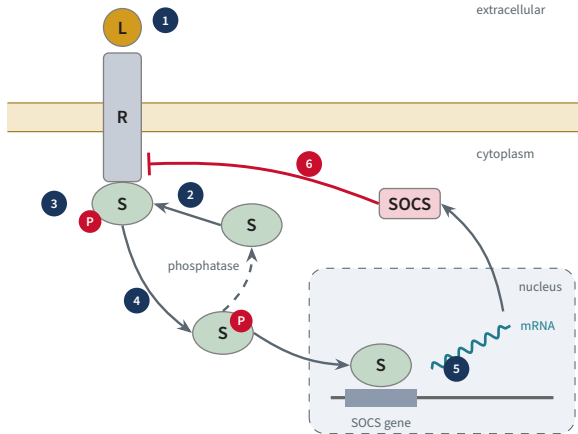
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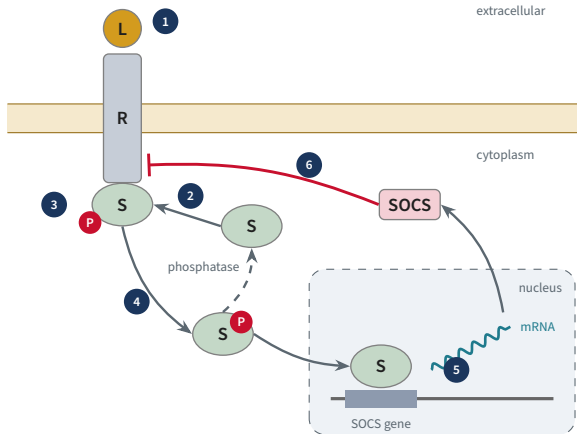
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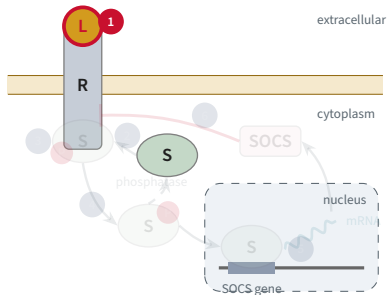
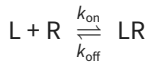
Each step adds reactions to the previous model.
The files are `models/stat_step1_binding.bngl`
through `stat_step6_feedback.bngl`.



Part 3

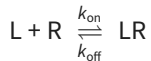
Step 1: Ligand–receptor binding

Binding is a reversible pair of reactions



Units: k_{on} in $\mu\text{M}^{-1}\text{min}^{-1}$, k_{off} in min^{-1}

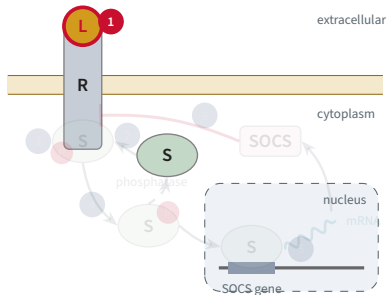
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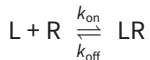
binding: $v_1 = k_{\text{on}} [L] [R]$

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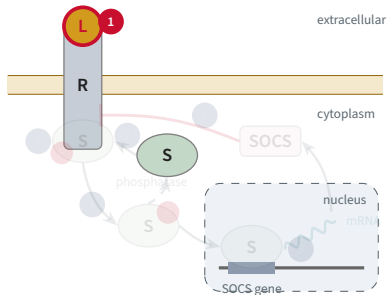
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Law of mass action

The rate of an elementary reaction is proportional to the product of the concentrations of its reactants. The proportionality constant is the **rate constant**.



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Each reaction contributes to the rate of change of its species

	v_1 (bind)	v_2 (unbind)
L	-1	+1
R	-1	+1
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Reactants are consumed ($-$), products are created ($+$).

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Ordinary differential equations (ODEs) follow mechanically from the list of reactions and their rate laws.

The stoichiometry matrix gives a compact general form

Collect the concentrations in a vector $\mathbf{X}$ and the reaction rates in $\mathbf{v}$:

$$\frac{d\mathbf{X}}{dt} = \mathbf{S} \mathbf{v}(\mathbf{X}), \quad \frac{d}{dt} \begin{pmatrix} [L] \\ [R] \\ [LR] \end{pmatrix} = \underbrace{\begin{pmatrix} -1 & +1 \\ -1 & +1 \\ +1 & -1 \end{pmatrix}}_{\mathbf{S}} \underbrace{\begin{pmatrix} k_{\text{on}}[L][R] \\ k_{\text{off}}[LR] \end{pmatrix}}_{\mathbf{v}(\mathbf{X})}$$

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- ▶ $\mathbf{S}$ has one row per species and one column per reaction.
- ▶ This form holds for *any* reaction network; only $\mathbf{S}$ and $\mathbf{v}$ change.

Conservation laws and equilibrium

Conservation. Binding only moves molecules between free and bound forms:

$$[L] + [LR] = L_0$$

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Small models can be solved by hand. Real signaling networks cannot, so we solve the ODEs numerically.

ODEs for reaction networks are solved numerically

Simplest idea (forward Euler): take small time steps

Δt ,

$$\mathbf{X}(t + \Delta t) \approx \mathbf{X}(t) + \Delta t \mathbf{S} \mathbf{v}(\mathbf{X}(t))$$

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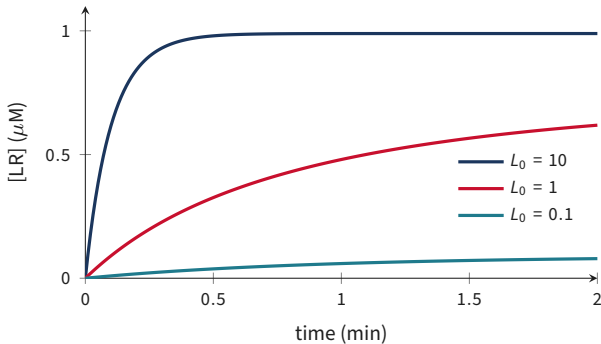
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In BioNetGen

```
simulate({method=>"ode", ...})  
uses CVODE.
```

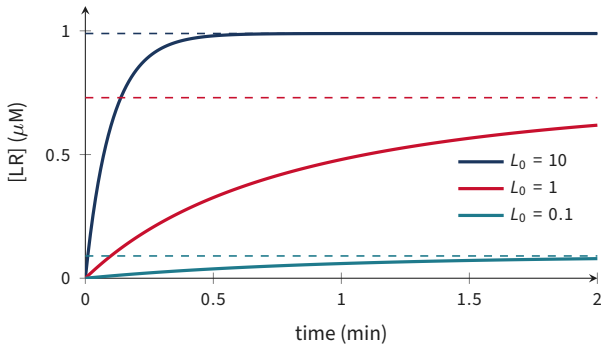
In Python, `model.setup_simulator()` returns a `libRoadRunner` simulator for the same model.

Binding approaches equilibrium faster and further at higher ligand



$$R_0 = 1 \mu\text{M}, k_{\text{on}} = 1 \mu\text{M}^{-1}\text{min}^{-1}, \\ k_{\text{off}} = 0.1 \text{ min}^{-1} (K_D = 0.1 \mu\text{M}).$$

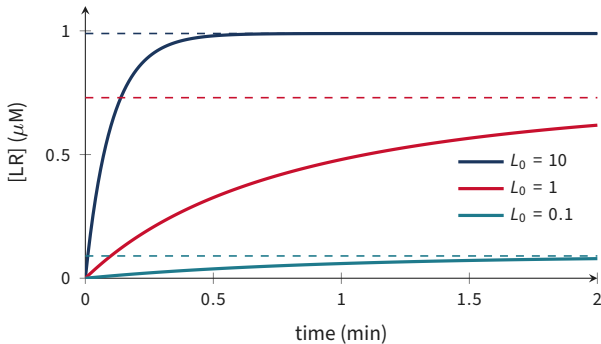
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The relaxation rate is $k_{\text{on}}([L] + [R]) + k_{\text{off}}$ near equilibrium, so higher ligand also means a faster response.

The same model in BNGL

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begin model

begin parameters
  L0      10      # ligand (uM)
  R0      1       # receptor (uM)
  kon     1       # binding (1/uM/min)
  koff    0.1     # unbinding (1/min)
end parameters

begin molecule types
  L
  R
  LR
end molecule types

begin species
  L    L0
  R    R0
end species
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  Molecules Lfree  L
  Molecules Rfree  R
  Molecules LR     LR
end observables

begin reaction rules
  # Ligand binds receptor
  LRbind: L + R <=> LR kon, koff
end reaction rules

end model

generate_network({overwrite=>1})
simulate({method=>"ode", t_end=>2, n_steps=>200})
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- ▶ **reaction rules:** $\leftrightarrow$ is reversible; forward and reverse rate constants

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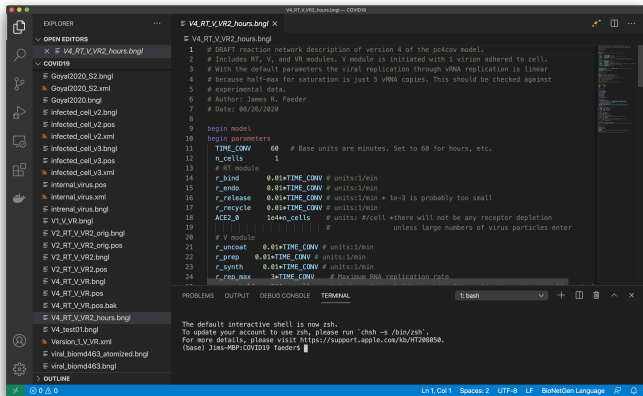
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  # Ligand binds receptor
  LRbind: L + R <-> LR kon, koff
end reaction rules

end model

generate_network({overwrite=>1})
simulate({method=>"ode", t_end=>2, n_steps=>200})
```

- ▶ **observables:** quantities to output
- ▶ **reaction rules:** $\leftrightarrow$ is reversible; forward and reverse rate constants
- ▶ **actions** after `end model`: generate the network, then simulate

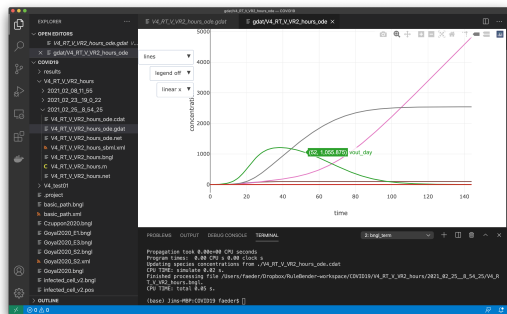
Running a model in VS Code



- ▶ Open a .bngl file
- ▶ Click the **run** button in the editor toolbar
- ▶ Results go to
BNG-results/<model>/
<timestamp>/

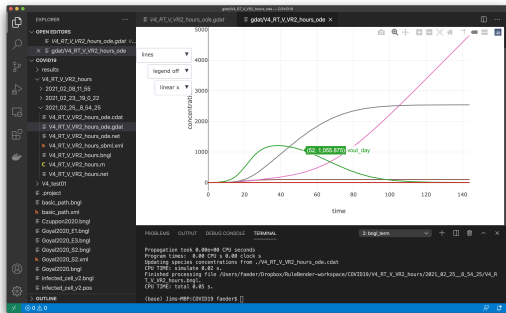
VS Code extension: Ali Sinan Saglam

Interactive plotting and Python notebooks

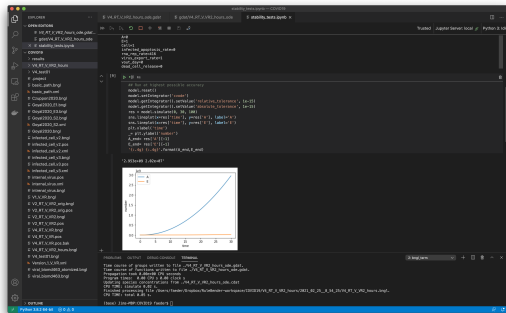


Built-in plotting of .gdat output

Interactive plotting and Python notebooks



Built-in plotting of .gdat output



Deeper analysis in Jupyter with PyBioNetGen +
libRoadRunner

BNG Playground: BioNetGen in your browser

- Write, simulate, and analyze BNGL models in a web browser: nothing to install



ruleworld.github.io/bngplayground

BNG Playground: BioNetGen in your browser

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BNG Playground: BioNetGen in your browser

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- ▶ Share a model as a link
- ▶ Useful for trying the course models on any computer: paste a file from `models/` and run it

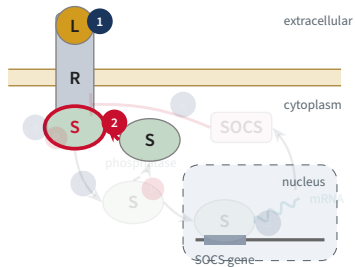


ruleworld.github.io/bngplayground

Part 4

Building the signaling circuit in BNGL

Step 2: STAT binds the ligand-bound receptor



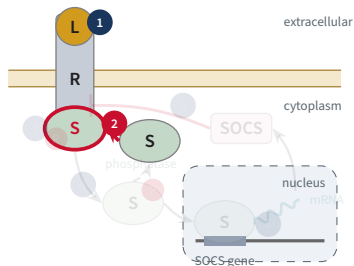
STAT can bind *only* the ligand-bound receptor LR, never free R. This is how ligand binding **enables** STAT recruitment.

Step 2: STAT binds the ligand-bound receptor



New lines in `stat_step2_recruit.bngl`:

```
begin parameters
  S0      10    # STAT (uM)
  kSon    1     # STAT binding (1/uM/min)
  kSoff   1     # STAT unbinding (1/min)
end parameters
begin molecule types
  S
  LRS
end molecule types
begin species
  S S0
end species
begin reaction rules
  Sbind: LR + S <-> LRS kSon, kSoff
end reaction rules
```



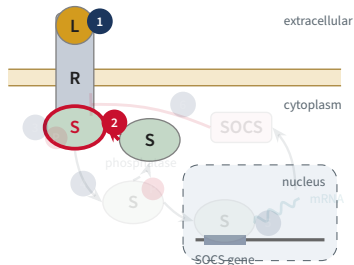
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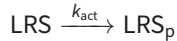
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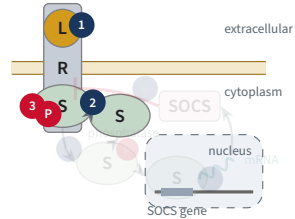
STAT can bind *only* the ligand-bound receptor LR, never free R. This is how ligand binding **enables** STAT recruitment.

Every new species needs a molecule type; species present at $t = 0$ also need an initial amount.

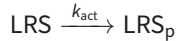
Step 3: receptor-bound STAT is activated



A receptor-associated kinase (JAK) phosphorylates the docked STAT: a **first-order**, irreversible reaction with $v = k_{\text{act}}[\text{LRS}]$.



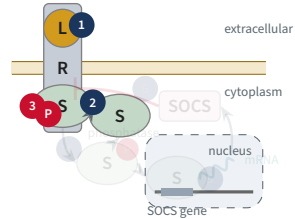
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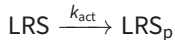
A receptor-associated kinase (JAK) phosphorylates the docked STAT: a **first-order**, irreversible reaction with $v = k_{\text{act}}[\text{LRS}]$.

```
kact 1 # STAT activation (1/min)
LRSp
Sact: LRS -> LRSp kact
```

one new line each in parameters, molecule types, and reaction rules



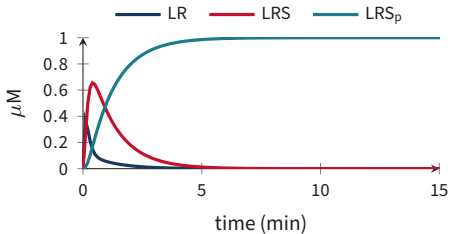
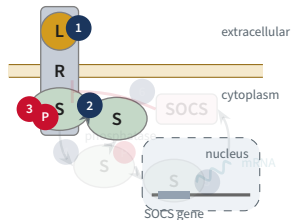
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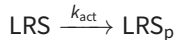
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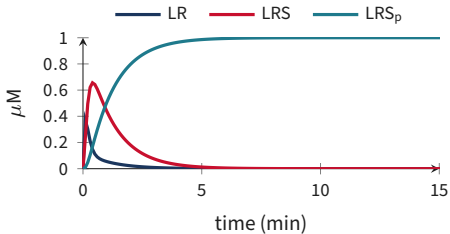
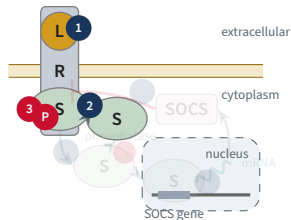


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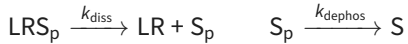
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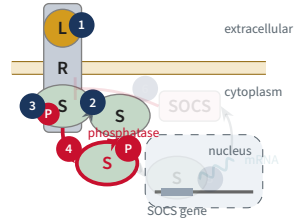
Problem: pSTAT never leaves the receptor. Every receptor ends up stuck as LRS_p .



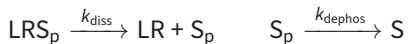
Step 4: activated STAT dissociates from the receptor



Releasing pSTAT frees the receptor to activate another STAT. A phosphatase returns free pSTAT to the inactive pool.

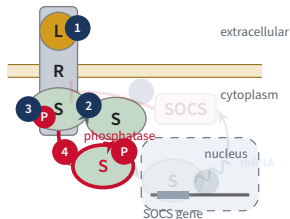


Step 4: activated STAT dissociates from the receptor

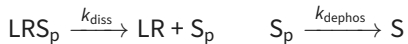


Releasing pSTAT frees the receptor to activate another STAT. A phosphatase returns free pSTAT to the inactive pool.

```
kdiss    1    # pSTAT release (1/min)
kdephos  0.1  # dephosphorylation (1/min)
Sp
Srelease: LRSp -> LR + Sp kdiss
Sdephos:  Sp -> S kdephos
```

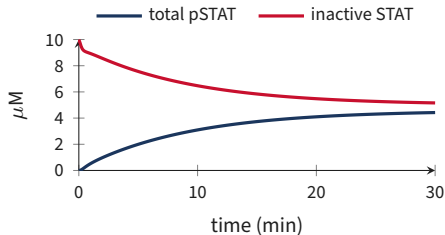
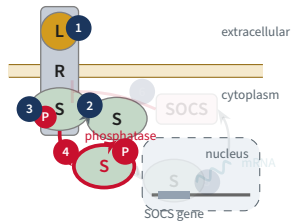


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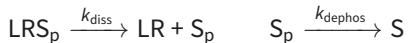


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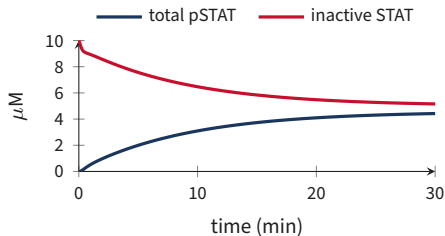
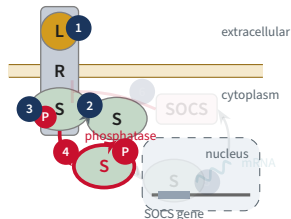
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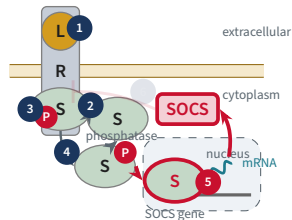
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pSTAT now reaches a steady state set by the balance of activation and dephosphorylation.



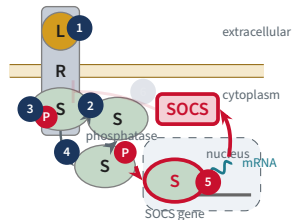
Step 5: pSTAT induces SOCS expression

```
# pSTAT binds the SOCS promoter
Gbind: Sp + G <-> GSp kGon, kGoff
# Bound promoter is transcribed
Transcribe: GSp -> GSp + mSOCS ktx
# mRNA degradation
mRNAdeg: mSOCS -> 0 kmdeg
# Translation
Translate: mSOCS -> mSOCS + SOCS kt1
# SOCS protein degradation
SOCSdeg: SOCS -> 0 kSOCSdeg
```



Step 5: pSTAT induces SOCS expression

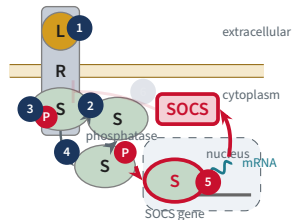
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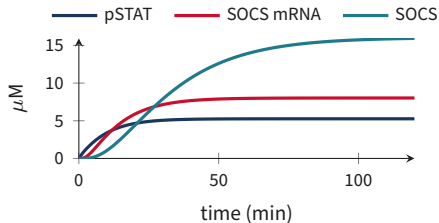
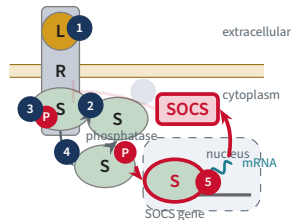


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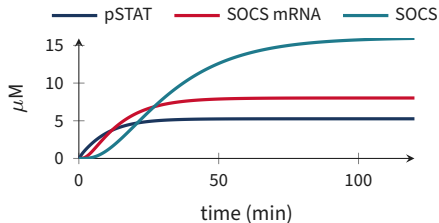
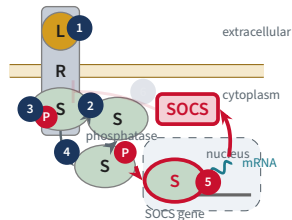


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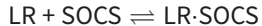
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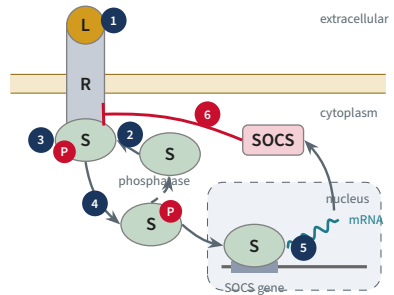
SOCS accumulates, but it has no target yet, so pSTAT is unaffected.



Step 6: SOCS binds the receptor and blocks STAT recruitment



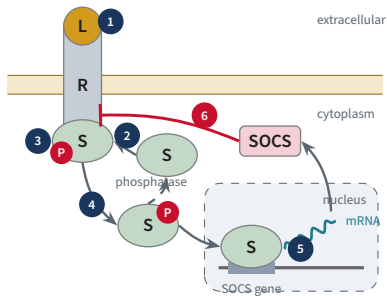
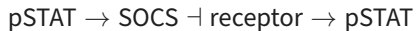
SOCS competes with STAT for the active receptor. LR·SOCS cannot recruit STAT, so pSTAT production drops.



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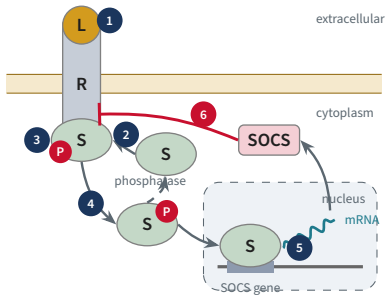
SOCS competes with STAT for the active receptor. LR·SOCS cannot recruit STAT, so pSTAT production drops. The loop is closed:



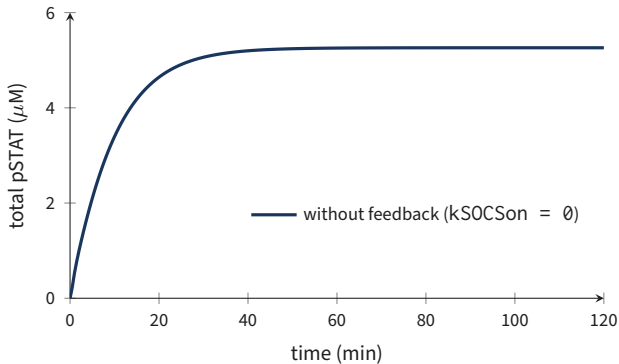
```

kSOCSon      1      # (1/uM/min)
kSOCSoFF     0.01   # (1/min)
LRSOCS
SOCSbind: LR + SOCS <=> LRSOCS kSOCSon, kSOCSoFF

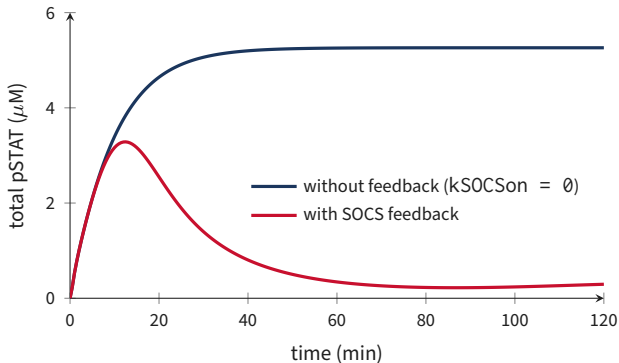
```



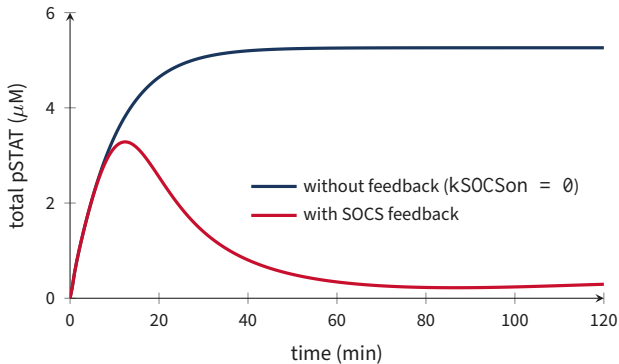
Negative feedback turns a sustained signal into a pulse



Negative feedback turns a sustained signal into a pulse

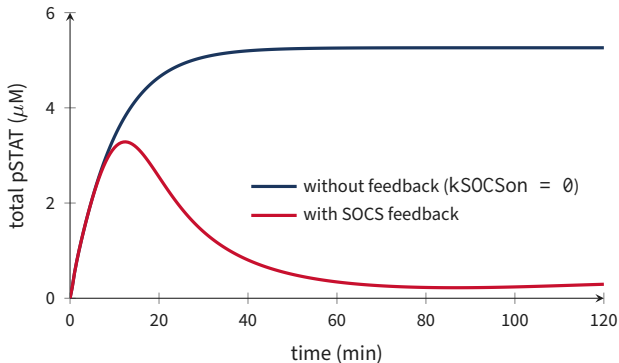


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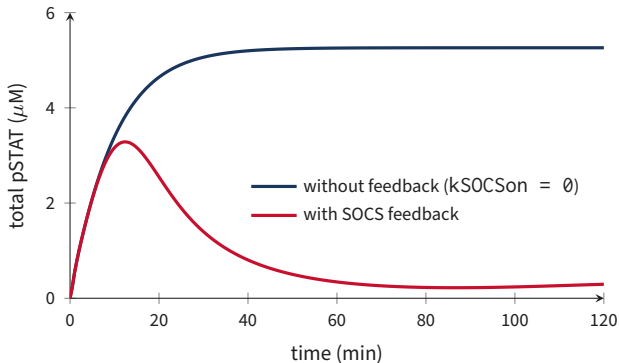
- Early response is identical: SOCS has not been made yet.

Negative feedback turns a sustained signal into a pulse



- ▶ Early response is identical: SOCS has not been made yet.
- ▶ As SOCS accumulates it sequesters receptors, and pSTAT falls to a low level despite continued ligand.

Negative feedback turns a sustained signal into a pulse



- ▶ Early response is identical: SOCS has not been made yet.
- ▶ As SOCS accumulates it sequesters receptors, and pSTAT falls to a low level despite continued ligand.
- ▶ The delay comes from transcription and translation.

Both curves come from `stat_step6_feedback.bng1`, which runs the model twice using `setParameter`.

The complete network

	reaction	rate
1	$L + R \rightleftharpoons LR$	k_{on}, k_{off}
2	$LR + S \rightleftharpoons LRS$	k_{Son}, k_{Soff}
3	$LRS \rightarrow LRS_p$	k_{act}
4	$LRS_p \rightarrow LR + S_p$	k_{diss}
	$S_p \rightarrow S$	k_{dephos}
5	$S_p + G \rightleftharpoons GS_p$	k_{Gon}, k_{Goff}
	$GS_p \rightarrow GS_p + mSOCS$	k_{tx}
	$mSOCS \rightarrow 0$	k_{mdeg}
	$mSOCS \rightarrow mSOCS + SOCS$	k_{tl}
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6	$LR + SOCS \rightleftharpoons LR \cdot SOCS$	$k_{SOCSon}, k_{SOCSoff}$

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- 11 lines of BNGL generate **15 reactions** (4 are reversible) among **12 species**.

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- ▶ 11 lines of BNGL generate **15 reactions** (4 are reversible) among **12 species**.
- ▶ **S** is a 12×15 matrix; BioNetGen builds it and the ODEs for you (see the .net file).
- ▶ Every species had to be named by hand: L, LR, LRS, LRS_p, LR·SOCS, ...

Part 5

Looking ahead

Naming every complex by hand does not scale

Our model made strong simplifying assumptions. Relax a few of them:

- ▶ SOCS can also bind receptors that already carry STAT (LRS, LRS_p)

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Each feature **multiplies** the number of distinct complexes, and each complex needs its own species name and its own copy of every reaction it takes part in.

Naming every complex by hand does not scale

Our model made strong simplifying assumptions. Relax a few of them:

- ▶ SOCS can also bind receptors that already carry STAT (LRS, LRS_p)
- ▶ Ligand can dissociate from *any* receptor complex, not only LR
- ▶ The receptor has two chains, each recruiting its own STAT
- ▶ STATs dimerize before entering the nucleus

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Module 2: represent molecules with internal structure (binding sites and states) and write rules that generate the network automatically.

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We return to stochastic simulation in a later module.

Try it yourself

Using the models in `models/`:

1. Run `stat_step1_binding.bngl` at several values of L_0 and check the equilibrium against the formula for $[LR]_{eq}$.
2. In `stat_step4_release.bngl`, set `kdephos` to 0. What happens to pSTAT, and why?
3. In `stat_step6_feedback.bngl`, vary `kt1` (translation rate). How do the peak and the late level of pSTAT change?
4. Add a reaction that lets SOCS bind LRS as well as LR. What new species do you need?

No local setup yet? Paste any of these models into [BNG Playground](https://ruleworld.github.io/bngplayground) (ruleworld.github.io/bngplayground) and run it in your browser.

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- ▶ Building the cytokine–STAT–SOCS circuit one step at a time shows how each mechanism contributes; negative feedback produces an adaptive pulse of pSTAT.
- ▶ Writing every complex as a separate species quickly becomes unmanageable. This motivates rule-based modeling, the subject of Module 2.